

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Output:

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priyanka@DESKTOP-H0ISO7A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 1$ uname -a
Linux DESKTOP-H0ISO7A 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
priyanka@DESKTOP-H0ISO7A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 1$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      8
On-line CPU(s) list:         0-7
Vendor ID:                   GenuineIntel
Model name:                  Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz
CPU family:                  6
Model:                      142
Thread(s) per core:         2
Core(s) per socket:         4
Socket(s):                   1
Stepping:                   10
BogoMIPS:                    4223.99
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mt
                             rr pge mca cmov pat pse36 clflush mmx fxsr sse
                             sse2 ss ht syscall nx pdpe1gb rdtscp lm arch_
                             perfmon rep_good nopl xtopology cpuid tsc_know
                             n_freq pni pclmulqdq vmx ssse3 fma cx16 pdcm p
                             cid sse4_1 sse4_2 movbe popcnt aes xsave avx f
                             16c rdrand hypervisor lahf_lm abm 3dnowprefetc
                             h pti ssbd ibrs ibpb stibp tpr_shadow ept vpid
                             ept_ad fsgsbase bti hle avx2 smep bmi2 erms
                             invpcid rtm rdseed adx smap clflushopt xsaveop
                             t xsavec xgetbv1 xsaves vnmi md_clear flush_l1
                             d arch_capabilities
Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                      128 KiB (4 instances)
  L1i:                      128 KiB (4 instances)
  L2:                       1 MiB (4 instances)
  L3:                       8 MiB (1 instance)
NUMA:
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NUMA node(s):          1
NUMA node0 CPU(s):     0-7
Vulnerabilities:
Gather data sampling:   Not affected
Ghostwrite:            Not affected
Indirect target selection: Mitigation; Aligned branch/return thunks
Itlb multihit:         KVM: Mitigation: Split huge pages
L1tf:                  Mitigation; PTE Inversion; VMX conditional cac
                        he flushes, SMT vulnerable
Mds:                   Mitigation; Clear CPU buffers; SMT Host state
                        unknown
Meltdown:              Mitigation; PTI
Mmio stale data:       Mitigation; Clear CPU buffers; SMT Host state
                        unknown
Old microcode:         Not affected
Reg file data sampling: Not affected
Retbleed:              Mitigation; IBRS
Spec rstack overflow:   Not affected
Spec store bypass:     Mitigation; Speculative Store Bypass disabled
                        via prctl
Spectre v1:            Mitigation; usercopy/swapgs barriers and __use
                        r pointer sanitization
Spectre v2:            Mitigation; IBRS; IBPB conditional; STIBP cond
                        itional; RSB filling; PBRSE-eIBRS Not affected
                        ; BHI SW loop, KVM SW loop
Srbds:                 Unknown: Dependent on hypervisor status
Tsa:                   Not affected
Tsx async abort:       Mitigation; Clear CPU buffers; SMT Host state
                        unknown
Vmscape:               Not affected
priyanka@DESKTOP-H0ISO7A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 1$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0     2G  0 disk [SWAP]
sdd   8:48   0     1T  0 disk /mnt/wslg/distro
/
priyanka@DESKTOP-H0ISO7A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 1$ ps
PID TTY          TIME CMD
337 pts/0        00:00:00 bash

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645 pts/0    00:00:00 ps
priyanka@DESKTOP-HOIS07A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 1$ top
top - 15:35:06 up 1 day, 3:55, 1 user, load average: 0.03, 0.17, 0.14
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0
MiB Mem : 7869.6 total, 7235.5 free, 516.5 used, 269.6 buff/cach
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7353.2 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+
    1 root        20   0   24320   15428   11624 S   0.0   0.2   0:01.75
    2 root        20   0    3180    2208    2072 S   0.0   0.0   0:00.03
    7 root        20   0    3180    2044    1940 S   0.0   0.0   0:00.00
   50 root        19  -1   50396   16788   15524 S   0.0   0.2   0:00.38
   85 systemd+    20   0   22416   14480   11988 S   0.0   0.2   0:00.24
   92 root        20   0   35252   12332    9228 S   0.0   0.2   0:00.48
  153 root        20   0    2888    1948    1820 S   0.0   0.0   0:00.16
  154 root        20   0    4472    3080    2832 S   0.0   0.0   0:00.01
  155 message+   20   0    8796    5404    4720 S   0.0   0.1   0:00.12
  162 root        20   0   44096   29836   14696 S   0.0   0.4   0:00.47
  166 root        20   0   18548    9436    8148 S   0.0   0.1   0:00.21
  168 root        20   0 1866032   14524   11848 S   0.0   0.2   0:00.28
  174 syslog      20   0   220548   5544    4608 S   0.0   0.1   0:00.15
  199 root        20   0    5492    2776    2588 S   0.0   0.0   0:00.02
  222 _chrony      20   0   20512    7932    7096 S   0.0   0.1   0:00.06
  236 _chrony      20   0   12096    2432    1792 S   0.0   0.0   0:00.00
  243 root        20   0  123460   32792   18180 S   0.0   0.4   0:00.31
  334 root        20   0    3184    1112     980 S   0.0   0.0   0:00.00
  335 root        20   0    3200    1248    1108 S   0.0   0.0   0:00.02
  337 priyanka    20   0    6268    5540    3864 S   0.0   0.1   0:00.10
  338 root        20   0    8644    5524    4684 S   0.0   0.1   0:00.03
  383 priyanka    20   0   22340   13596   10976 S   0.0   0.2   0:00.20
  387 priyanka    20   0   22912    4128    2128 S   0.0   0.1   0:00.00
  415 priyanka    20   0    6136    5408    3864 S   0.0   0.1   0:00.03
  655 priyanka    20   0    8172    6008    3908 R   0.0   0.1   0:00.00

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The Linux operating system acts as an interface between applications and hardware. Commands such as **lscpu**, **lsblk**, **ps** and **top** allow users to observe CPU, storage, and process information without directly interacting with the hardware. The kernel manages these resources and provides abstractions such as processes, files, memory, and device interfaces, making hardware easier and safer for applications to use.